

**Definition 18.1**

Let  $R$  be a commutative ring. An element  $a \neq 0$  of  $R$  is a *zero divisor* if there exists  $b \neq 0$  such that  $ab = 0$ .

**Definition 18.2**

An *integral domain* is a commutative ring with unity which has no zero divisors.

**Theorem 18.3**

Let  $R$  be an integral domain and  $a, b, c \in R$ . If  $a \neq 0$  and  $ab = ac$  then  $b = c$ .

**Definition 18.4**

Let  $R$  be a commutative ring with unity. An element  $a \in R$  is a *unit* if there exists  $b \in R$  such that  $ab = 1$ . In such case, we denote  $a^{-1} := b$ .

**Definition 18.5**

A *field* is a commutative ring with unity in which every non-zero element is a unit.

**Theorem 18.6**

Every field is an integral domain.

**Theorem 18.7**

A ring  $\mathbb{Z}_n$  is a field if and only if  $n$  is a prime number.

**Definition 18.8**

Let  $F$  be a field with unity  $1 \in F$ . The *characteristic* of  $F$  is the smallest positive integer  $n$  such that

$$\underbrace{1 + 1 + \dots + 1}_{n \text{ times}} = 0$$

denote such  $n$  by  $\chi(F)$ .

If such  $n$  does not exist, then  $\chi(F) = 0$

### Theorem 18.9

- 1) If  $F$  is a field then  $\chi(F)$  is either 0 or a prime number.
- 2) If  $F$  is a finite field and  $\chi(F) = p$  for some prime  $p$ , then  $F$  consists of  $p^n$  elements for some  $n \geq 1$ .

**Example:** Field with 9 elements.